

Sales Analysis Report

Project Title

Sales Analysis Using Python

Objective

The objective of this project is to analyze retail sales data to identify business trends, calculate key performance indicators (KPIs), determine top-performing products and regions, and generate visual insights for decision-making.

Tools and Technologies

- Python
- Pandas
- Matplotlib
- Visual Studio Code

Dataset Description

The dataset contains retail sales transactions recorded throughout the year 2025. Each record includes:

- Order ID
- Order Date
- Product
- Region
- Quantity
- Unit Price

Revenue for each transaction is calculated using:

$$\text{Revenue} = \text{Quantity} \times \text{Unit Price}$$

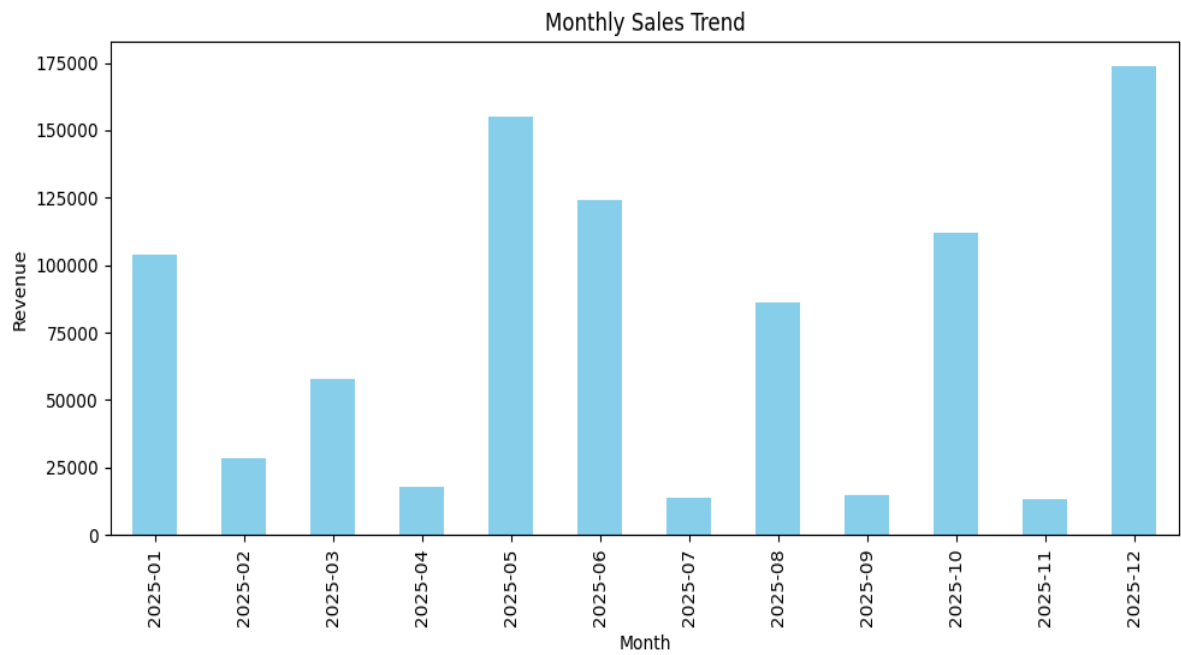
Key Performance Indicators (KPIs)

The analysis calculates:

- Total Revenue
- Total Orders
- Average Order Value
- Top Products by Revenue
- Top Regions by Revenue

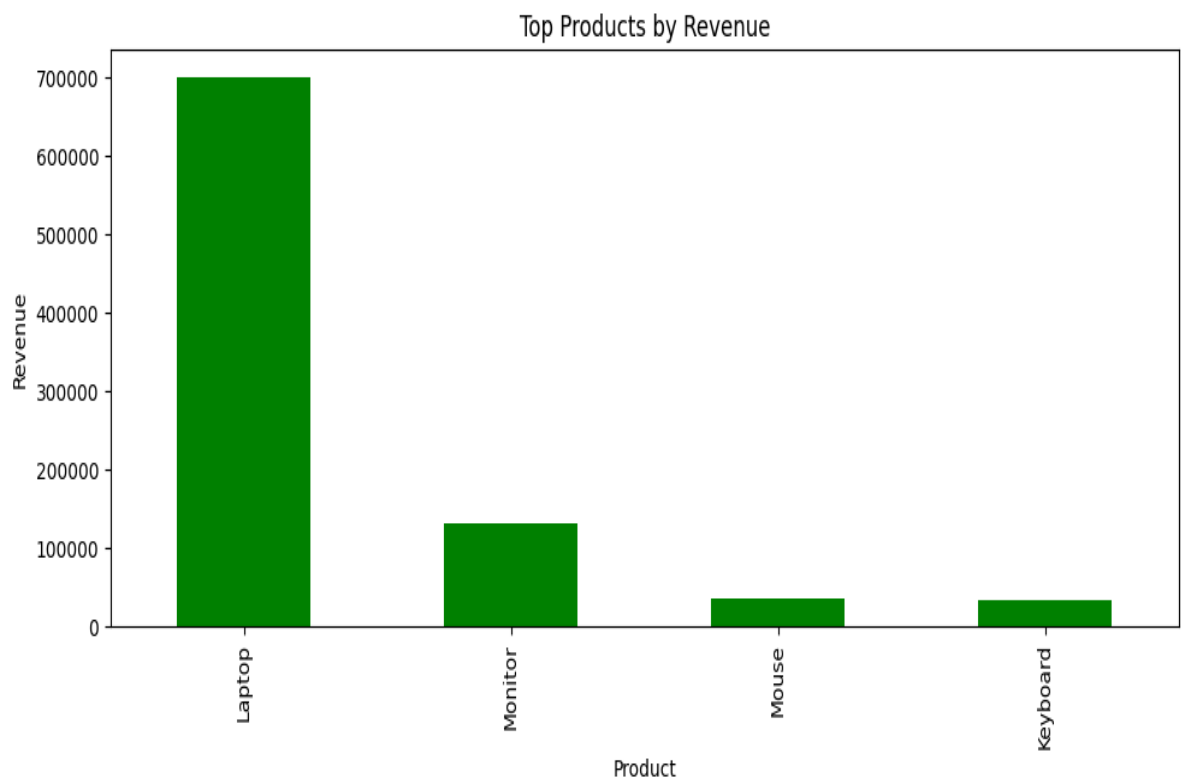
Sales Trend Analysis

Monthly sales were analyzed across multiple months to understand seasonal performance and overall revenue trends.



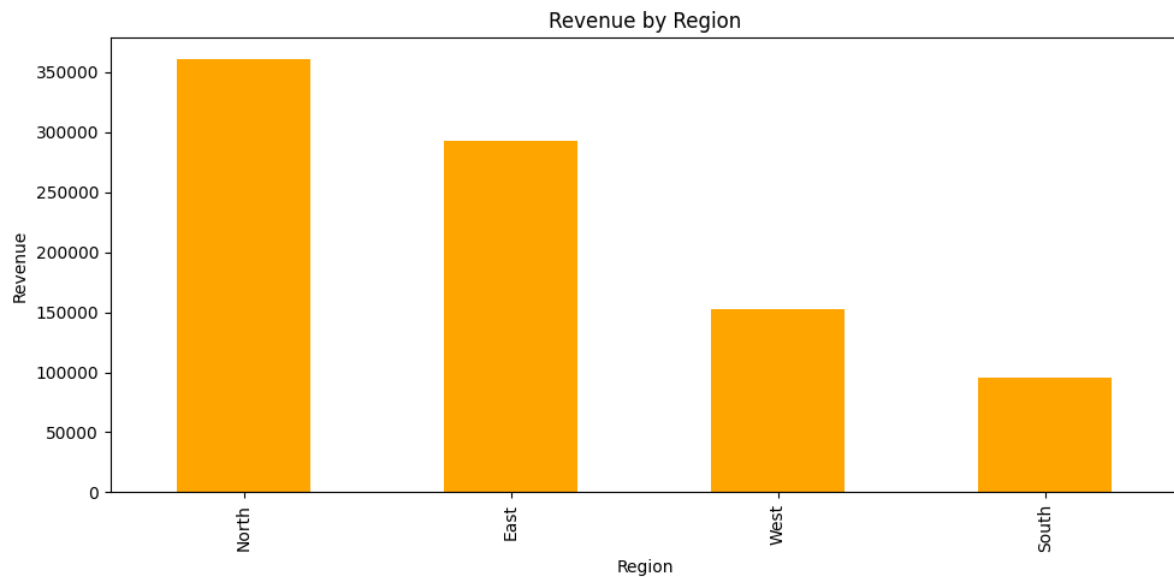
Top Products

The revenue generated by each product category was compared to identify the highest-performing products.



Regional Performance

Sales revenue was analyzed across different regions to determine the strongest markets.



Business Insights

1. Laptop sales contribute the largest share of overall revenue.
2. Revenue varies across regions, indicating differences in market performance.
3. Monthly sales trends help identify periods of stronger customer demand.

Recommendations

- Maintain sufficient inventory for high-performing products.
- Increase marketing efforts in regions with lower sales.
- Monitor monthly sales patterns to improve planning and forecasting.
- Continue using data-driven analysis to support business decisions.

Conclusion

This project demonstrates the use of Python for data cleaning, KPI calculation, exploratory data analysis, and visualization. The generated insights can help businesses improve inventory management, marketing strategies, and overall sales performance.